Clinical Transplantation

WILEY

Clinical TRANSPLANTATION
The Journal of Clinical and Translational Research

ORIGINAL ARTICLE

CT-Based Lung Size Matching in Delayed Chest Closure for Systemic Sclerosis Lung Transplantation

Jatin Singh[1] | Xin Meng[1] | Joseph K. Leader[1] | John Ryan[2] | Ernest G. Chan[2] | Norihisa Shigemura[2] | Chadi A. Hage[3] | Pablo G. Sanchez[4] | Jiantao Pu[1,5,6]

[1]Department of Radiology, University of Pittsburgh, Pittsburgh, Pennsylvania, USA | [2]Division of Lung Transplant and Lung Failure, Department of Cardiothoracic Surgery, University of Pittsburgh School of Medicine, Pittsburgh, Pennsylvania, USA | [3]Division of Pulmonary Medicine and Critical Care, Department of Medicine, University of Pittsburgh School of Medicine, Pittsburgh, Pennsylvania, USA | [4]Department of Surgery, Section of Thoracic Surgery, University of Chicago, Chicago, Illinois, USA | [5]Department of Bioengineering, University of Pittsburgh, Pittsburgh, Pennsylvania, USA | [6]Department of Ophthalmology, University of Pittsburgh, Pittsburgh, Pennsylvania, USA

Correspondence: Jiantao Pu (jip13@pitt.edu)

Received: 12 September 2024 | **Revised:** 27 October 2024 | **Accepted:** 11 November 2024

Funding: This work is supported in part by research grants from the National Institutes of Health (NIH) (R01HL174570 and R01CA237277) and UPMC Hillman Developmental Pilot Program.

Keywords: delayed chest closure | lung transplant | scleroderma | systemic sclerosis

ABSTRACT

Purpose: Delayed chest closure (DCC) during lung transplantation (LTx) is a controversial surgical approach that lacks research in systemic sclerosis (SSc) patients. We investigated outcomes, clinical risk factors, and CT-based lung size-matching parameters associated with DCC in SSc recipients.

Methods: This retrospective study included 92 SSc recipients (age 51 years ± 10, 56/92 (61.0%) females) who underwent bilateral LTx between 2007 and 2020. Of the recipients, 34.8% (32/92) underwent DCC. Recipient lung and chest cavity volumes were automatically computed from CT imaging using deep learning algorithms. Survival between groups was compared using Kaplan–Meier analysis. Multivariate logistic regression was used to identify risk factors and predict DCC occurrence using preoperative variables.

Results: Recipients who underwent DCC had longer *total vent duration* ($p = 0.001$), more use of *postoperative mechanical support* ($p = 0.001$), longer *ICU length of stay* ($p = 0.008$), and lower incidence of *pneumonia* post-operation ($p = 0.031$). No significant difference in survival was observed between DCC and PCC recipients at 30 days ($p = 0.713$), 90 days ($p = 0.267$), 1 year ($p = 0.941$), and 5 years ($p = 0.651$). Clinical risk factors for DCC included *BMI > 30 kg/m*^{2} ($p = 0.009$), *tracheostomy* ($p = 0.002$), *atrial fibrillation* ($p = 0.012$), decreased *preoperative FEV1/FVC* ($p = 0.013$), and *previous chest operation* ($p = 0.020$). Two CT-based measurements of lung matching were significantly associated with DCC occurrence ($p = 0.021$ and 0.050). The regression model achieved a mean AUC of 0.82 (0.70, 0.94) in retrospectively predicting DCC occurrence.

Conclusion: SSc recipients undergoing DCC have similar survival rates but experience more complications than PCC recipients. Clinical risk factors and CT-based size matching can be leveraged to predict DCC pre-transplant.

Abbreviations: AUC, area under the curve; CI, confidence interval; DCC, delayed chest closure; ILD, interstitial lung disease; LTx, lung transplant; ML, machine learning; PAH, pulmonary arterial hypertension; PCC, primary chest closure; SSc, systemic sclerosis.

© 2024 John Wiley & Sons A/S. Published by John Wiley & Sons Ltd.

1 | Introduction

Systemic sclerosis (SSc) is an idiopathic, autoimmune disorder characterized by diffuse fibrosis of connective tissue throughout the body [1]. Among its manifestations, pulmonary complications are the leading cause of mortality in SSc patients. These complications primarily occur in the form of interstitial lung disease (ILD), pulmonary arterial hypertension (PAH), or a combination of both [2, 3]. Currently, lung transplantation (LTx) is the only effective treatment in the management of end-stage lung disease. However, LTx in patients with Ssc is a rare procedure, accounting for only around 1% of total LTx procedure volume [2]. This low percentage can be attributed to the complex pathogenesis of the disease, which restricts the number of centers capable of conducting LTx in SSc patients. Therefore, there is a significant gap in research related to the complications and outcomes associated with LTx in SSc patients [2–4], which needs to be addressed to improve LTx outcomes.

Delayed chest closure (DCC), also known as delayed sternal closure, is a surgical strategy employed in instances of hemodynamic instability, significant intraoperative injuries, oversized grafts, pulmonary edema, and coagulopathy to mitigate primary graft dysfunction (PGD) and irreversible lung injury [5, 6]. In the case of oversized donor grafts, donor lung size reduction or even lobectomy can be utilized as an alternative approach [7]. DCC can be similarly employed to address lung-size mismatching among other complications. However, the use of DCC remains controversial, and specific risk factors and outcomes for DCC are not well-documented [5, 6, 8–12]. In a meta-analysis of DCC conducted by Chen et al. [5], DCC was associated with a higher risk of PGD compared to primary chest closure (PCC), along with a lower 30-day and 5-year survival. However, in the largest case series on DCC, Shigemura et al. [6] found that DCC and strict protocol-driven postoperative management effectively prevented compression-related injury and could improve survival and functional status. The conflicting evidence suggests a need for more research on DCC. Additionally, to our knowledge, the implications of DCC use have not been explicitly studied in the context of LTx for SSc patients.

During the pre-LTx assessment of potential LTx patients, chest CT scans are routinely acquired to assess the extent of lung disease. Recent advances in image analysis software have opened the door for automatic quantitative image analysis from CT imaging [13]. Iyer et al. [14] used CT-derived body composition features, such as adipose tissue and muscle metrics, to augment the prediction of esophageal cancer survival, achieving an AUC of 0.861 to predict 5-year survival. Automated CT analysis holds the potential to reveal valuable information that could influence or even predict transplant complications and outcomes. However, few LTx studies have leveraged CT-derived features in their analyses. Barbosa et al. [15] found that models incorporating software-generated CT metrics performed better than radiologists scoring in predicting chronic lung allograft dysfunction (CLAD) with a maximum AUC of 0.817. In another study, Horie et al. [16] focused on recipients experiencing a decline in FEV1, discovering that quantitative lung density metrics obtained from post-LTx CT scans were significant predictors (AUC = 0.74) of CLAD. To our knowledge, only Sukhavasi et al. [17] have leveraged CT-derived measurements while analyzing DCC versus PCC in heart transplant recipients. They used estimated cardiac mass generated from machine learning models to discriminate recipients who underwent DCC over PCC and achieved an AUC score of 0.804. Currently, recipients at our institution are matched to donor lungs using the European Respiratory Society's predicted total lung capacity (pTLC) equations [18] that consider sex and height. However, inferential equations have inherent limitations, especially in restrictive lung diseases, prompting the necessity for individualized lung sizing. For this reason, we aim to study CT-based volumetric analysis of the recipient lungs and chest cavity to improve the current standard of size matching, a key factor in DCC.

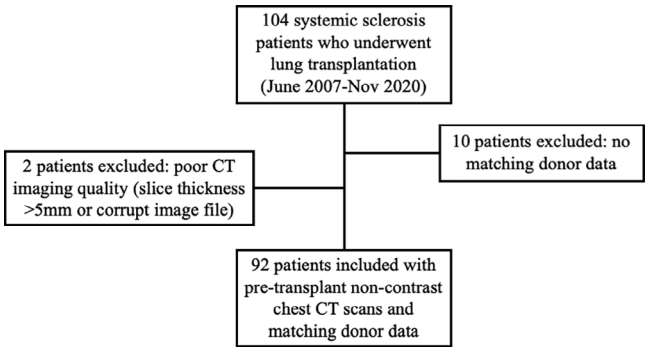

FIGURE 1 | Inclusion flowchart for the study cohort.

To shed light on DCC in SSc-related LTx, we investigated preoperative recipient characteristics, intra-operative surgical features, and postoperative outcomes. Deep learning algorithms were used to automatically segment and quantify lung and chest cavity features from standard pre-LTx chest CT imaging. These features provided an additional tool to estimate lung allograft size matching. Finally, a multivariate logistic regression model was fitted to the data to predict DCC occurrence using only pre-LTx features.

2 | Materials and Methods

2.1 | Study Cohort

We collected data on 92 SSc recipients who underwent bilateral LTx at the University of Pittsburgh Medical Center from 2007 to 2020. Patient inclusion is described in Figure 1. Among these recipients, 34.8% (32/92) underwent DCC during surgery. Graft size mismatch was the leading indication for DCC, representing more than half of recipients (n = 18). Following graft size mismatch, coagulopathy (n = 14), reperfusion injury (n = 4), and hemodynamics (n = 2) were documented as the indications for DCC. Two cases had no indication specified, while seven cases had multiple indications (Table SA). All patients underwent a transsternal bilateral thoracotomy incision (clamshell incision) for surgical access.

The protocol for chest closure begins with closure at the time of transplantation for the vast majority of patients. The chest is then reapproximated once hemostasis is achieved. Clinical variables

13990012, 2024, 12, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ctr.70041 by Lund University, Wiley Online Library on [22/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

including blood pressure (systolic, diastolic, and mean arterial pressure), pulmonary pressures (systolic, diastolic, and mean), and peak airway pressures are compared to pre-approximation values to ensure they have not changed dramatically. After this, the patient is checked for hemodynamic changes to ensure the patient can tolerate chest closure. If the recipient exhibits signs of instability, DCC is initiated.

Pre-LTx non-contrast chest CT scans from the patients were performed on various multi-slice [19] CT scanners (Optima CT660 [GE Healthcare], Lightspeed 16 [GE Healthcare], Lightspeed Plus [GE Healthcare], and Lightspeed VCT [GE Healthcare]). Recipients were positioned feet-first supine. Scan parameters include tube voltage 120 kV and tube current 302 ± 160 mAs. The CT images were reconstructed using the GE "lung" kernel as a matrix of 512 × 512 with an image thickness ranging from 0.625 to 4 mm. The series with the maximum number of CT image slices were selected for the analyses. This study was approved by the University of Pittsburgh Institutional Review Board (IRB #: 24030161).

2.2 | CT Image Features

In-house three-dimensional (3-D) convolutional neural networks (CNN) algorithms [20–23] were used to segment and quantify recipient lungs and chest cavity sizes from CT imaging. Specifically, the Unet model with a grid-based three-dimensional patch sampling operation [23] was applied to segment the structures. These algorithms have been validated across multiple studies [19, 23, 24] and performed reliably on various CT protocols and scanners. Due to the unavailability of donor CT imaging, we were unable to generate these metrics from the donors.

2.3 | Clinical Features

Subject (recipient and donor) characteristics pertaining to preoperative, intraoperative, and postoperative variables were collected from electronic medical records. Features with missing values, lack of variance, or unreadable data were excluded from the analyses. Donor characteristics included age, gender, BMI, height, race, smoking status, and pTLC. The underlying diagnosis of ILD or PAH in the SSc recipients was compared between the DCC and PCC recipients. Blood transfusion requirements during surgery were expressed as the sum of the number of units of red blood cells, packs of fresh frozen plasma, bags of platelets, and units of cryoprecipitate. The pTLC for the donors was estimated due to a lack of donor TLC data using Equation (1) where gender = 0 for female and 1 for male and race = 0 for non-African American and 1 for African American [25].

$$pTLC = (-630.819 + (967.100 * \text{ gender}) + (25.197 \text{ height}) - (713.838 * \text{ race}) + (15.103 * \text{ age}) \quad (1)$$

These pTLC measurements were then used to generate three ratios estimating donor-recipient lung size matching. The $D_{\text{pTLC}}/R_{\text{pTLC}}$ Ratio combined both pTLC ratios generated using Equation (1) between donors (D) and recipients (R). The $D_{\text{pTLC}}/R_{\text{CT Lung Vol}}$ Ratio used the CT-derived lung volume measurement for recipients and the $D_{\text{pTLC}}/R_{\text{CT Chest Cavity Vol}}$ Ratio used the CT-derived chest cavity measurement.

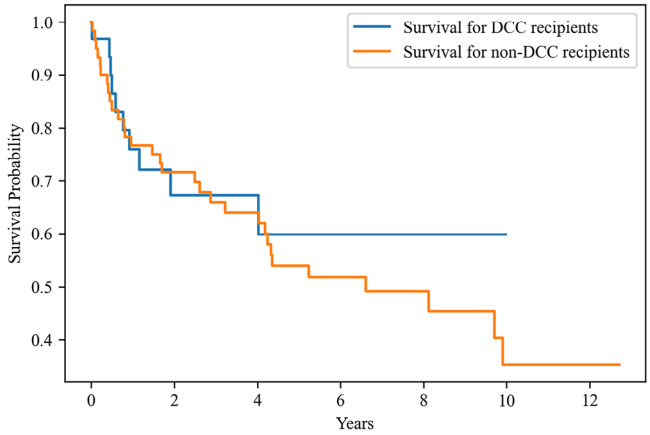

FIGURE 2 | Kaplan–Meier plot comparing survival distributions of DCC and non-DCC LTx recipients.

2.4 | Statistical Analysis

All categorical variables were one-hot encoded to enable comparisons. Variables with less than 5% missing data were imputed using mean or mode imputation, depending on the variable type. Variables with more than 5% missing data were excluded. Univariate analysis was performed using Fisher's exact test for categorical variables and the Mann–Whitney U test for continuous variables. A summary of the univariate results is provided since 130 variables were tested in total; the complete analysis is available in Table SB. Kaplan–Meier analysis was utilized to compare survival between the DCC and PCC groups at 30-day, 90-day, 1-year, and 5-year intervals. For subjects surviving past the cutoff date, right censoring was applied. A p value of less than 0.05 was considered statistically significant.

Before multivariate analysis, continuous features were z-score standardized. Features with a variance inflation factor (VIF) greater than 5 were eliminated to minimize collinearity. Using a multivariate logistic regression model, preoperative features were iteratively selected via a backward stepwise elimination strategy using a p value cutoff of 0.05. The final model was trained on the data using stratified 5-fold cross-validation to ensure robustness and minimize overfitting. A grid search was performed to tune model hyperparameters. Model performance was evaluated using accuracy and the area under the curve (AUC). All analyses were conducted using Python 3.9

3 | Results

3.1 | Outcomes

The Kaplan–Meier median survival time for the cohort was 8.12 years. Mortality rates in the DCC subset were similar to those in the PCC subset at all tested time points (Figure 2), with no significant difference in survival between the DCC and PCC

13990012, 2024, 12, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ctr.70041 by Lund University, Wiley Online Library on [22/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

groups at 30 days ($p = 0.713$), 90 days ($p = 0.267$), 1 year ($p = 0.941$), and 5 years ($p = 0.651$) (Table 1,2).

Post-operation, recipients who underwent DCC had longer *total vent duration ($p = 0.001$)*, increased use of *postoperative mechanical support ($p = 0.001$)*, and longer *ICU length of stay ($p = 0.008$)*. Notably, recipients who underwent DCC had a significantly lower incidence of *pneumonia* post-operation ($p = 0.031$) (Table 1).

3.2 | Clinical Risk Factors

We observed that SSc recipients with BMI > 30 kg/m^2 ($p = 0.009$), tracheostomy ($p = 0.002$), atrial fibrillation ($p = 0.012$), decreased preoperative FEV1/FVC ($p = 0.013$), and previous chest operation ($p = 0.020$) underwent DCC at a higher rate. During surgery, recipients undergoing DCC more often required VAEMO support ($p = 0.009$) and wedge resections ($p = 0.010$) (Tables 1 and 2).

3.3 | Lung Size Matching

From the three estimation ratios of lung allograft matching, two ratios ($D_{\text{pTLC}}/R_{\text{CT Lung Vol}}$ Ratio, $p = 0.021$; $D_{\text{pTLC}}/R_{\text{CT Chest Cavity Vol}}$ Ratio, $p = 0.050$) were significantly associated with DCC occurrence. In the $D_{\text{pTLC}}/R_{\text{CT Lung Vol}}$ Ratio, recipients with a larger mean ratio were associated with DCC, potentially indicating that larger mismatches necessitated DCC use. In the D_{pTLC} / $R_{\text{CT Chest Cavity Vol}}$ Ratio, a lower mean ratio was associated with PCC, potentially indicating that larger recipient chest cavities were better suited for donor lung allografts (Table 2).

3.4 | Multivariate Analysis

Six features were selected after stepwise feature elimination, including Previous Chest Operation ($\beta = 2.08$, $p = 0.004$), Atrial Fibrillation ($\beta = 1.82$, $p = 0.006$), BMI > 30.0 kg/m^2 ($\beta = 1.92$, $p = 0.011$), ACE Inhibitor Therapy ($\beta = -2.87$, $p = 0.017$), Tracheostomy ($\beta = 1.36$, $p = 0.018$), and Donor Age ($\beta = -0.67$, $p = 0.045$) (Table 3). Based on the hyperparameter grid search, the multivariate logistic regression model was fitted using the L2 penalty, regularization penalty of 0.1, and the "liblinear" solver. The model via 5-fold cross-validation achieved a mean accuracy of 0.73 (0.60, 0.86) and a mean AUC of 0.82 (95% CI: 0.70, 0.94) in predicting DCC occurrence pre-LTx (Figure 3).

4 | Discussion

DCC is a controversial surgical strategy employed in the management of numerous complications during LTx [5, 6, 8–12]. The existing understanding of the risk factors associated with the need to utilize DCC and the outcomes following DCC remains inadequate, particularly in the context of autoimmune disease-related LTx like SSc. We investigated the risk factors and outcomes for DCC recipients with the addition of CT thoracic metrics to predict lung allograft size matching. Our results show that SSc recipients who undergo DCC have similar short- and long-term survival rates to SSc recipients who undergo PCC, which reflects the findings of other studies [9–12]. Our multivariate

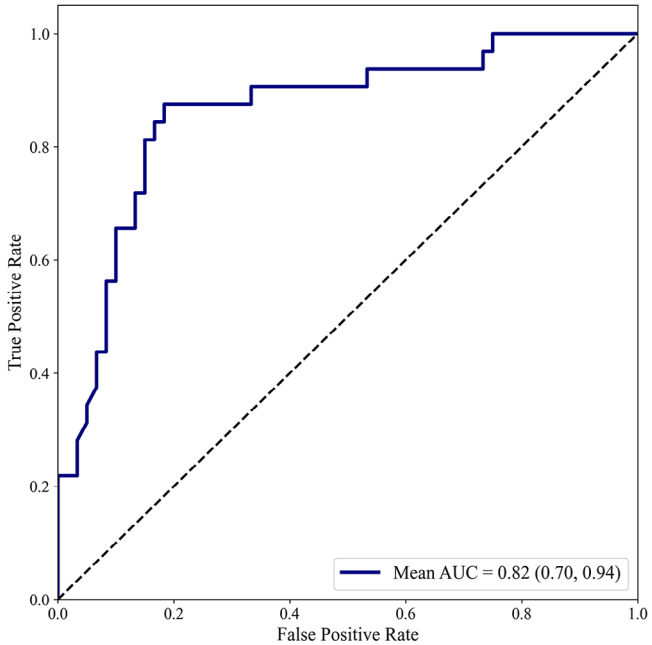

FIGURE 3 | ROC plot of multivariate logistic regression model performance on DCC prediction using pre-operative features.

logistic regression model, using only six pre-LTx features, could discriminate which recipients would undergo DCC with an AUC of 0.82 (95% CI: 0.70, 0.94).

We observed that certain clinical features were risk factors for DCC occurrence. Recipients classified as obese based on BMI (>30 kg/m^2) were more likely to undergo DCC. This finding reflects past studies on DCC [9–12] and the correlation between high BMI and surgical complications. The association of DCC with recipients with tracheostomies, atrial fibrillation, decreased FEV1/FVC, and previous chest operations likely reflects general poor health and progressed lung disease, requiring alternative therapies like DCC. There was no observed association between SSc type, (ILD vs. PAH) and DCC ($p = 0.053$). In multivariate analysis, donor age ($\beta = -0.67$, $p = 0.045$) emerged as an additional significant feature, indicating that lungs from younger donors resulted in less DCC utilization. During operation, recipients undergoing DCC required VAECMO support and wedge resection more often, which reflects the use of DCC in hemodynamically unstable patients and for oversized lung allografts. Finally, recipients undergoing DCC faced more adverse outcomes than PCC recipients including longer total vent duration, ICU length of stay, and increased reliance on postoperative mechanical support.

Lung allograft size mismatch is an established negative prognostic marker in LTx, contributing to increased risk of PGD and mortality [26]. Currently, pTLC estimations are used to match recipients to donor lungs. The simplified nature of these pTLC equations cannot accurately account for variations in populations and may overestimate lung volume sizes in connective-tissue lung disease. CT-based volumetric analysis provides individualized and accurate lung size measurements that may be superior to inferential equations. The two ratios we generated using CT-derived lung metrics were significantly different among

13990012, 2024, 12, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ctr.70041 by Lund University, Wiley Online Library on [22/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1 | Recipient and donor demographics and delayed chest closure ($n = 92$).

Characteristic	Statistic	DCC ($n = 32$)	PCC ($n = 60$)	p value
Age (years)				
Mean ± SD	51 ± 10	51 ± 8	51 ± 11	0.990
Gender				
Male	35/92 (38.0%)	9/32 (28.1%)	26/60 (43.3%)	0.124
Female	56/92 (61.0%)	23/32 (71.9%)	33/60 (55.0%)	
BMI (kg/m^2)				
Mean ± SD	25.1 ± 4.9	26.3 ± 5.3	24.4 ± 4.6	0.097
<18.5	7/92 (7.6%)	2/32 (6.2%)	5/60 (8.3%)	1.000
18.5–24.9	40/92 (43.5%)	13/32 (40.6%)	27/60 (45.0%)	0.826
25.0–29.9	28/92 (30.4%)	6/32 (18.8%)	22/60 (36.7%)	0.097
>30.0	17/92 (18.5%)	11/32 (34.4%)	6/60 (10.0%)	**0.009**
Height (cm)				
Mean ± SD	168.4 ± 9.1	167.4 ± 8.5	168.9 ± 9.5	0.4147
Race				
White	68/92 (73.9%)	22/32 (68.8%)	46/60 (76.7%)	0.460
African American	20/92 (21.7%)	9/32 (28.1%)	11/60 (18.3%)	0.299
Smoking status				
Former	39/92 (42.4%)	12/32 (37.5%)	27/60 (45.0%)	0.515
>20 pack-years	17/92 (18.5%)	6/32 (18.8%)	11/60 (18.3%)	1.000
Recipient pTLC (L)				
Mean ± SD	4.6 ± 0.8	4.4 ± 0.7	4.7 ± 0.8	0.096
Donor age (years)				
Mean ± SD	35 ± 15	35 ± 14	36 ± 15	0.840
Donor gender				
Male	31/92 (33.7%)	12/32 (37.5%)	19/60 (31.7%)	0.646
Female	61/92 (66.3%)	20/32 (62.5%)	41/60 (68.3%)	
Donor BMI (kg/m^2)				
Mean ± SD	26.2 ± 5.8	25.9 ± 5.4	26.5 ± 6.0	0.664
Donor height (cm)				
Mean ± SD	164.8 ± 9.8	163.1 ± 8.5	165.8 ± 10.4	0.084
Donor race				
White	61/92 (66.3%)	23/32 (71.9%)	38/60 (63.3%)	0.491
African American	23/92 (25.0%)	6/32 (18.8%)	17/60 (28.3%)	0.449
Donor smoking status				
>20 pack-years	11/92 (12.0%)	3/32 (9.4%)	8/60 (13.3%)	0.742
Donor pTLC (L)				
Mean ± SD	4.2 ± 0.7	4.2 ± 0.6	4.2 ± 0.7	0.831

Note: The bold values indicate significant values.

Abbreviations: DCC = delayed chest closure, PCC = primary chest closure, pTLC = predicted total lung capacity (based on age, gender, race, and height), SSc = systemic sclerosis, SD = standard deviation.

13990012, 2024, 12, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ctr.70041 by Lund University, Wiley Online Library on [22/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2 | Univariate association of features with DCC ($n = 92$).

Characteristic	Statistic	DCC ($n = 32$)	PCC ($n = 60$)	p value
		CT lung volumetric features		
$D_{pTLC}/R_{CT\ Lung\ Vol}$ ratio	1.5 ± 0.6	1.7 ± 0.5	1.5 ± 0.6	**0.0210**
$D_{pTLC}/R_{CT\ Chest\ Cavity\ Vol}$ ratio	0.95 ± 0.29	1.00 ± 0.26	0.91 ± 0.31	**0.0496**
CT Lung Vol (L)	3.0 ± 1.0	2.7 ± 0.8	3.2 ± 1.1	0.0525
D_{pTLC}/R_{pTLC} ratio	0.9 ± 0.2	1.0 ± 0.2	0.9 ± 0.2	0.0645
CT chest cavity vol (L)	4.8 ± 1.4	4.4 ± 1.1	5.0 ± 1.5	0.0889
		Preoperative features		
Tracheostomy	42/92 (45.7%)	22/32 (68.8%)	20/60 (33.3%)	**0.002**
Atrial fibrillation	30/92 (32.6%)	16/32 (50.0%)	14/60 (23.3%)	**0.012**
FEV1/FVC	0.81 ± 0.21	0.76 ± 0.13	0.83 ± 0.24	**0.013**
Previous chest operation	21/92 (22.8%)	12/32 (37.5%)	9/60 (15.0%)	**0.020**
ACE inhibitor therapy	13/92 (14.1%)	1/32 (3.1%)	12/60 (20.0%)	**0.030**
Ssc-PAH	27/92 (29.3%)	5/32 (15.6%)	22/60 (36.7%)	0.053
Ssc-ILD	65/92 (70.7%)	27/32 (84.4%)	38/60 (63.3%)	0.053
mPAP (mmHg)	36.3 ± 11.1	38.4 ± 11.6	35.3 ± 10.8	0.221
LAS	61.6 ± 18.8	64.8 ± 18.6	59.9 ± 18.8	0.236
LV ejection fraction	57.3 ± 8.6	55.5 ± 10.3	58.2 ± 7.4	0.287
		Intra-operative features		
VAECMO support	29/92 (31.5%)	16/32 (50.0%)	13/60 (21.7%)	**0.009**
Wedge resection	22/92 (23.9%)	13/32 (40.6%)	9/60 (15.0%)	**0.010**
Surgery time (h)	8.1 ± 1.6	8.5 ± 1.7	8.0 ± 1.5	0.131
Transfusion units	13.4 ± 22.1	15.2 ± 27.6	12.4 ± 18.6	0.165
ECMO bridge to transplant	9/92 (9.8%)	5/32 (15.6%)	4/60 (6.7%)	0.268
CPB support	43/92 (46.7%)	16/32 (50.0%)	27/60 (45.0%)	0.667
MV bridge to transplant	11/92 (12.0%)	4/32 (12.5%)	7/60 (11.7%)	1.000
		Postoperative outcomes		
Total vent duration (h)	422.7 ± 762.1	594.2 ± 1016.3	331.2 ± 573.3	**0.001**
Postop mechanical support	25/92 (27.2%)	16/32 (50.0%)	9/60 (15.0%)	**0.001**
ICU LOS	20.8 ± 31.7	28.0 ± 42.5	16.9 ± 23.7	**0.008**
Postop pneumonia	49/92 (53.3%)	12/32 (37.5%)	37/60 (61.7%)	**0.031**
Hospital LOS	41.4 ± 44.0	49.8 ± 63.3	36.9 ± 28.7	0.498
30-Day mortality	2/92 (2.2%)	1/32 (3.1%)	1/60 (1.7%)	0.713
90-Day mortality	7/92 (7.6%)	1/32 (3.1%)	6/60 (10.0%)	0.267
1-Year mortality	22/92 (23.9%)	8/32 (25.0%)	14/60 (23.3%)	0.941
5-Year mortality	41/92 (44.6%)	13/32 (40.6%)	28/60 (46.7%)	0.651

Note: The bold values indicate significant values.

Abbreviations: CT Vol = CT-derived lung volume, D = donor, DCC = delayed chest closure, ILD = interstitial lung disease, LAS = lung allocation score, LOS = length of stay, LTx = lung transplant, LV = left ventricular, PAH = pulmonary arterial hypertension, PCC = primary chest closure, pTLC = predicted lung capacity, R = recipient, SSc = systemic sclerosis, VAECMO = veno-arterial ECMO.

13990012, 2024, 12, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ctr.70041 by Lund University, Wiley Online Library on [22/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 3 | Multivariate logistic regression analysis results ($n = 92$).

Characteristic	Log-odds	95% CI	*p* value
Previous chest operation	2.08	(0.66, 3.50)	0.004
Atrial fibrillation	1.82	(0.53, 3.11)	0.006
BMI > 30.0 kg/m^2	1.92	(0.45, 3.39)	0.011
ACE inhibitor therapy	−2.87	(−5.23, −0.50)	0.017
Tracheostomy	1.36	(0.24, 2.48)	0.018
Donor age (years)	−0.67	(−1.32, −0.01)	0.045

the DCC and PCC groups. On average, recipients with larger $D_{\text{pTLC}}/R_{\text{CT Lung Vol}}$ Ratios were more likely to undergo DCC ($p = 0.021$). We theorized that this represented a larger mismatch between donor and recipient lungs, a mismatch not covered using traditional pTLC matching. Additionally, recipients with larger $D_{\text{pTLC}}/R_{\text{CT Chest Cavity Vol}}$ Ratios were also more likely to undergo DCC on average ($p = 0.049$). We theorized that better allograft matching was achieved when the CT-derived chest cavity volume was larger than the donor pTLC, which would facilitate an easier sternal closure.

This study has several limitations. First, the sample size ($n = 92$) was small and derived from a single center, which may affect the generalizability of the findings. However, this dataset represents the largest group of LTx in SSc patients with available CT scans studied to date. The rarity of centers performing LTx in SSc patients and the low incidence of SSc make it challenging to find an external validation cohort. Second, there was missing clinical data, such as incomplete information on primary graft dysfunction (PGD) and CLAD, which are the leading causes of mortality in LTx. Consequently, this data was not included in the study. Third, the CT scans were performed over 13 years using different protocols and scanners, which led to variations in image quality. Our 3D-CNN body composition algorithms have been extensively validated against many CT scans acquired using various protocols [9, 26], which minimizes the impact of data heterogeneity. Nonetheless, future testing is necessary.

5 | Conclusion

Lung transplantation in SSc patients is a complex procedure that would benefit from an advanced understanding of complications pre-LTx to guide appropriate clinical decisions. Our study sheds light on the preoperative risk factors for DCC in SSc patients, as well as the intra-operative and postoperative outcomes associated with DCC use. Survival between DCC and PCC recipients was not significantly different; however, DCC was associated with more postoperative complications. Additionally, the use of CT-based lung size matching better differentiated between recipients who underwent DCC or PCC, indicating that CT-based ratios may be more discriminate and accurate than traditional pTLC estimations. Multivariate logistic regression models trained on the collected pre-LTx data could accurately differentiate recipients who underwent DCC and PCC.

Author Contributions

Jatin Singh participated in the writing of the paper, performance of the research, and data analysis. Xin Meng participated in the writing of the paper and data analysis. Joseph K. Leader participated in the writing of the paper. John Ryan participated in the writing of the paper. Ernest G. Chan participated in data collection and data analysis. Norihisa Shigemura participated in the writing of the paper. Chadi A. Hage participated in the performance of the research and data collection. Pablo G. Sanchez participated in the performance of the research and data collection. Jiantao Pu participated in the writing of the paper, performance of the research, and data analysis.

Acknowledgments

This work is supported in part by research grants from the National Institutes of Health (NIH) (R01HL174570 and R01CA237277) and UPMC Hillman Developmental Pilot Program.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

1. C. P. Denton and D. Khanna, "Systemic Sclerosis," *Lancet* 390, no. 10103 (2017): 1685–1699, https://doi.org/10.1016/S0140-6736(17)30933-9.

2. A. Minalyan, L. Gabrielyan, S. Khanal, B. Basyal, and C. Derk, "Systemic Sclerosis: Current State and Survival After Lung Transplantation," *Cureus* 13, no. 1 (2021): e12797, https://doi.org/10.7759/cureus.12797.

3. P. Pradere, I. Tudorache, J. Magnusson, et al., "Lung Transplantation for Scleroderma Lung Disease: An International, Multicenter, Observational Cohort Study," *Journal of Heart and Lung Transplantation* 37, no. 7 (2018): 903–911, https://doi.org/10.1016/j.healun.2018.03.003.

4. R. Jablonski, J. Dematte, and S. Bhorade, "Lung Transplantation in Scleroderma: Recent Advances and Lessons," *Current Opinion in Rheumatology* 30, no. 6 (2018): 562–569, https://doi.org/10.1097/BOR.0000000000000546.

5. C. Chen, Q. Zheng, D. Wu, Y. Song, and G. Xu, "Review of Outcomes of Delayed Chest Closure Following Lung Transplantation: A Meta-Analysis," *Journal of Cardiothoracic Surgery* 17, no. 1 (2022): 122, https://doi.org/10.1186/s13019-022-01868-w.

6. N. Shigemura, Y. Orhan, J. K. Bhama, et al., "Delayed Chest Closure After Lung Transplantation: Techniques, Outcomes, and Strategies," *Journal of Heart and Lung Transplantation* 33, no. 7 (2014): 741–748, https://doi.org/10.1016/j.healun.2014.03.003.

7. N. Shigemura, C. Bermudez, B. G. Hattler, et al., "Impact of Graft Volume Reduction for Oversized Grafts After Lung Transplantation on Outcome in Recipients With End-Stage Restrictive Pulmonary Diseases," *Journal of Heart and Lung Transplantation* 28, no. 2 (2009): 130–134, https://doi.org/10.1016/j.healun.2008.11.003.

8. P. R. Aguilar, B. C. Bemiss, C. Witt, et al., "Impact of Delayed Chest Closure on Surgical Site Infection After Lung Transplantation," *Annals of Thoracic Surgery* 104, no. 4 (2017): 1208–1214, https://doi.org/10.1016/j.athoracsur.2017.05.023.

9. S. D. Force, D. L. Miller, A. Pelaez, et al., "Outcomes of Delayed Chest Closure After Bilateral Lung Transplantation," *Annals of Thoracic Surgery* 81, no. 6 (2006): 2020–2024, https://doi.org/10.1016/j.athoracsur.2006.01.050. discussion 2024-5.

13990012, 2024, 12, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ctr.70041 by Lund University, Wiley Online Library on [22/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

10. T. Hirama, M. Akiba, M. Ui, et al., "Long-Term Survival and Clinical Outcomes of Delayed Chest Closure Following Lung Transplantation," *Surgery Today* 54, no. 10 (2024): 1138–1145, https://doi.org/10.1007/s00595-024-02821-1.

11. S. Rafiroiu, H. Hassouna, U. Ahmad, et al., "Consequences of Delayed Chest Closure During Lung Transplantation," *Annals of Thoracic Surgery* 109, no. 1 (2020): 277–284, https://doi.org/10.1016/j.athoracsur.2019.08.016.

12. A. Yeginsu, A. E. Tasci, M. Vayvada, et al., "Delayed Chest Closure for Oversized Lung Allograft in Lung Transplantation: A Retrospective Analysis From Turkey," *Brazilian Journal of Cardiovascular Surgery* 36, no. 6 (2021): 760–768, https://doi.org/10.21470/1678-9741-2020-0299.

13. T. Elhakim, K. Trinh, A. Mansur, C. Bridge, and D. Daye, "Role of Machine Learning-Based CT Body Composition in Risk Prediction and Prognostication: Current State and Future Directions," *Diagnostics (Basel)* 13, no. 5 (2023): 968, https://doi.org/10.3390/diagnostics13050968.

14. K. Iyer, C. A. Beeche, N. S. Gezer, et al., "CT-Derived Body Composition Is a Predictor of Survival After Esophagectomy," *Journal of Clinical Medicine* 12, no. 6 (2023): 2106, https://doi.org/10.3390/jcm12062106.

15. E. M. Barbosa Jr., S. Simpson, J. C. Lee, N. Tustison, J. Gee, and H. Shou, "Multivariate Modeling Using Quantitative CT Metrics May Improve Accuracy of Diagnosis of Bronchiolitis Obliterans Syndrome After Lung Transplantation," *Computers in Biology and Medicine* 89 (2017): 275–281, https://doi.org/10.1016/j.compbiomed.2017.08.016.

16. M. Horie, L. Levy, C. Houbois, et al., "Lung Density Analysis Using Quantitative Chest CT for Early Prediction of Chronic Lung Allograft Dysfunction," *Transplantation* 103, no. 12 (2019): 2645–2653, https://doi.org/10.1097/TP.0000000000002771.

17. A. Sukhavasi, D. Ahmad, M. Austin, et al., "Utility of Recipient Cardiothoracic Ratio in Predicting Delayed Chest Closure After Heart Transplantation," *Thoracic and Cardiovascular Surgeon* 72, no. 4 (2024): 253–260, https://doi.org/10.1055/a-2015-1507.

18. J. P. Ouwens, T. W. van der Mark, W. van der Bij, A. Geertsma, W. J. de Boer, and G. H. Koeter, "Size Matching in Lung Transplantation Using Predicted Total Lung Capacity," *European Respiratory Journal* 20, no. 6 (2002): 1419–1422, https://doi.org/10.1183/09031936.02.00294402.

19. L. Pu, S. F. Ashraf, N. S. Gezer, et al., "Estimating 3-D Whole-Body Composition from a Chest CT Scan," *Medical Physics* 49, no. 11 (2022): 7108–7117, https://doi.org/10.1002/mp.15821.

20. J. Pu, J. K. Leader, J. Sechrist, et al., "Automated Identification of Pulmonary Arteries and Veins Depicted in Non-Contrast Chest CT Scans," *Medical Image Analysis* 77 (2022): 102367, https://doi.org/10.1016/j.media.2022.102367.

21. J. Pu, J. Roos, C. A. Yi, S. Napel, G. D. Rubin, and D. S. Paik, "Adaptive Border Marching Algorithm: Automatic Lung Segmentation on Chest CT Images," *Computerized Medical Imaging and Graphics* 32, no. 6 (2008): 452–462, https://doi.org/10.1016/j.compmedimag.2008.04.005.

22. L. Pu, J. K. Leader, A. Ali, Z. Gent, and D. Wilson, "Predicting Left/Right Lung Volumes, Thoracic Cavity Volume, and Heart Volume From Subject Demographics to Improve Lung Transplant," *Journal of Medical Imaging* 10, no. 5 (2023): 051806.

23. L. Pu, N. S. Gezer, S. F. Ashraf, I. Ocak, D. E. Dresser, and R. Dhupar, "Automated Segmentation of Five Different Body Tissues on Computed Tomography Using Deep Learning," *Medical Physics* 50, no. 1 (2023): 178–191, https://doi.org/10.1002/mp.15932.

24. J. Pu, J. K. Leader, J. Sechrist, et al., "Automated Identification of Pulmonary Arteries and Veins Depicted in Non-Contrast Chest CT Scans," *Medical Image Analysis* 77 (2022): 102367, https://doi.org/10.1016/j.media.2022.102367.

25. J. A. Konheim, Z. N. Kon, C. Pasrija, et al., "Predictive Equations for Lung Volumes from Computed Tomography for Size Matching in Pulmonary Transplantation," *Journal of Thoracic and Cardiovascular Surgery* 151, no. 4 (2016): 1163–1169.e1, https://doi.org/10.1016/j.jtcvs.2015.10.051.

26. A. Tuba and Y. Ali, "Size Matching in Lung Transplantation: A Narrative Review," *Archives of Organ Transplantation* 5, no. 1 (2020): 001–005, https://doi.org/10.17352/2640-7973.000014.

Supporting Information

Additional supporting information can be found online in the Supporting Information section.

13990012, 2024, 12, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ctr.70041 by Lund University, Wiley Online Library on [22/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License